

Shane Woloszyn

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EDUCATION

Purdue University

West Lafayette, IN

Bachelor of Science in Computer Science and Economics

Aug. 2025 – May 2028

- **Activities:** John Martinson Honors College • BuildPurdue • Purdue Magnetics Lab • Sigma Alpha Epsilon – Recruitment Chair • Certificate in Entrepreneurship and Innovation
- **Coursework:** Discrete Math, Programming in C, Linear Algebra, OOP, Computer Architecture, Data Structures & Algorithms, Multivariate Calculus

EXPERIENCE

Software Engineering Intern

May 2026 – Present

Liberty Mutual Insurance Group

Portsmouth, NH

- Contributing to Twilio Flex contact center platform supporting 20k+ concurrent agents across call handling, compliance workflows, and analytics
- Re-architecting the platform's event fanout service around AWS EventBridge and serverless Lambda, migrating to an event-driven architecture for cleaner service decoupling and improved horizontal scalability
- Building backend logic and TypeScript/React UI for an FCRA disclosure system handling 4% of eligible calls at a 5M+ monthly call center
- Collaborating with senior engineers in an Agile environment, participating in code review and sprint planning across a cross-functional team

Co-Founder and Lead Developer

Dec. 2025 – Present

Liora

West Lafayette, IN

- Founded and led engineering for a hotel management platform; owned architecture, hiring, and product direction from day one
- Built initial product in Java with Spark and Twilio, shipping prototypes for customer demos within the first month; selected for VentureX accelerator with \$5k funding

Independent Researcher

June 2024 – Feb. 2025

University of New England

Biddeford, ME

- Designed a CGAN in Python/PyTorch to generate synthetic chest X-ray images for medical education; validated outputs against ground-truth CT data
- Presented findings at the Summer Undergraduate Research Symposium under Dr. Sylvain Jaume

PROJECTS

C++ Limit Order Book Matching Engine | C++, pybind11, pytest, GitHub

- Single-threaded engine hitting 2.43M orders/sec at p50 300ns; price-time priority matching, partial fills, and IOC/FOK/Market order types
- O(1) cancellation via hashmap-iterator locator; exposed to Python via pybind11 and validated with a 29-test pytest suite

Crypto Arbitrage Trading Dashboard | Python, Streamlit, CCXT

- Real-time arbitrage detector pulling live order book data from 5+ exchanges via CCXT and flagging fee-aware profit opportunities
- Backtested strategies across thousands of simulated paper trades, tracking cumulative and per-trade returns

File Loading Operating System | C, x86 Assembly, Git

- Built a bootloader and kernel supporting file read/write on a simulated RAM disk, with VGA text output and PS/2 keyboard input via direct hardware port I/O
- Implemented real-to-protected mode transition on x86, including GDT setup and segment register configuration

TECHNICAL SKILLS

Languages: Python, Java, C, C++, TypeScript, JavaScript, x86 Assembly

Frameworks & Tools: React, PyTorch, NumPy, Unity, Spark, Git, Unix/Linux, Twilio, REST APIs

Concepts: Machine learning, distributed systems, object-oriented design, Agile/Scrum, data analytics, algorithmic trading